basic bound / sub. G

markov bound $\longrightarrow$ $\boxed{X > 0}$

$$\mathbb{P}[X > t] \leq \frac{\mathbb{E}[X]}{t}$$

$$\mathbb{P}\left[|X-\mu|^{k} > t^{k}\right] \leq \frac{\mathbb{E}\left[|X-\mu|^{k}\right]}{t}$$

$k=2 \longrightarrow$ chebyshev

$$f(x) = e^{\lambda x} \longrightarrow \mathbb{P}\left(e^{\lambda x} > e^{\lambda t}\right) \leq \frac{\mathbb{E}\left[e^{\lambda x}\right]}{e^{\lambda t}}$$

chernoff

$$\mathbb{P}[X > t] = \int_0^\infty \mathbb{1}(x > t)\, p(x)\, dx = \mathbb{E}\left[\mathbb{1}(X > t)\right]$$

$$\frac{1}{t} \times \mathbb{E}\left[t \times \mathbb{1}(X > t)\right] \leq \frac{1}{t}\mathbb{E}\left[X \times \mathbb{1}\{X > t\}\right]$$

$$\leq \frac{1}{t}\mathbb{E}[X]$$

$X \in \{0, t\}$ $\longrightarrow$ bound tight

$$\underbrace{(X - \mathbb{E}[X])^r}_{} \in \underbrace{\{0, t\}}_{}$$

$$X = \begin{cases} r\sqrt{t} & p \\ \sqrt{t} & 1 - rp \\ 0 & p \end{cases}$$

$$\mathbb{P}(X > t) \leq \inf_{k} \frac{\mathbb{E}[X^k]}{t^k}$$

$$\mathbb{P}(X > t) \leq \inf_{\lambda} \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda t}} \qquad \lambda > 0$$

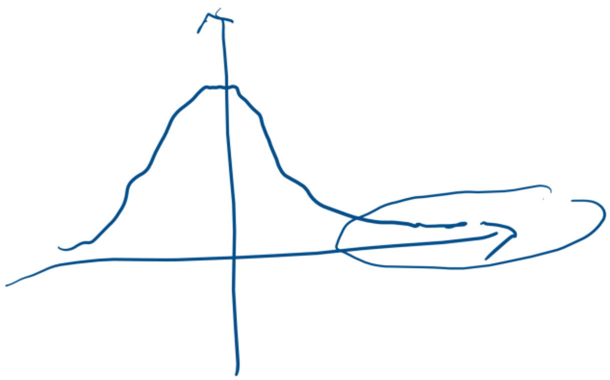
$$\mathbb{E}[e^{\lambda X}] = \mathbb{E}\sum_{n \geq 0}^{\infty} \frac{\lambda^n X^n}{n!} = \mathbb{E}\sum_{n \geq 0}^{\infty} \frac{\lambda^n t^n}{n!} \times \frac{X^n}{t^n}$$

$$\sum_{n \geq 0}^{\infty} \frac{\lambda^n t^n}{n!} \times \frac{\mathbb{E}[X^n]}{t^n} \geq \underbrace{\sum \frac{(\lambda t)^n}{n!}}_{} \inf_{k} \frac{\mathbb{E}[X^k]}{t^k}$$

$$= e^{\lambda t} \inf_x (\sim)$$

$$\boxed{\inf_k \frac{\mathbb{E}[X^k]}{t^k} \le \inf_\lambda e^{-\lambda t}\, \mathbb{E}[e^{\lambda X}]}$$

$X \sim N(\mu, \sigma^2)$ $\quad \underbrace{\mathbb{E}[e^{\lambda X}]} = e^{\lambda\mu + \frac{\lambda^2\sigma^2}{2}}$



Sub-Gaussian Dist.

iff $\exists\, \sigma > 0$ $\quad \mathbb{E}[e^{\lambda(X-\mu)}] \le e^{\frac{\lambda^2\sigma^2}{2}}$

Rad: $\{-1, 1\}$

$X \in [a, b]$ $\qquad \boxed{\sigma = \frac{(b-a)}{2}}$

$$\psi(\lambda) = \log \mathbb{E}\left[e^{\lambda X}\right]$$

$$\psi(0) = 0 \qquad \psi'(0) = \mu$$

$$\psi'(\lambda) = \frac{1}{\mathbb{E}\left[e^{\lambda X}\right]} \times \mathbb{E}\left[X e^{\lambda X}\right]$$

$$\psi'(0) = \frac{1}{1} \times \mathbb{E}\left[X \times 1\right] = \mu$$

$$\psi(\lambda) = \psi(0) + \psi'(0) \times \lambda + \psi''(\eta) \frac{\lambda^2}{2}$$

$$\eta \in [0, \lambda]$$

$$\psi'' \leq ?$$

$$\psi''(\lambda) = \frac{+\mathbb{E}\left[X^2 e^{\lambda X}\right] \times \mathbb{E}\left[e^{\lambda X}\right] - \mathbb{E}\left[X e^{\lambda X}\right]^2}{\mathbb{E}\left[e^{\lambda X}\right]^2}$$

$$= \frac{\mathbb{E}[X^2 e^{\lambda X}]}{\mathbb{E}[e^{\lambda X}]} - \left(\frac{\mathbb{E}[X e^{\lambda X}]}{\mathbb{E}[e^{\lambda X}]}\right)^2$$

$$\int x^2 \frac{e^{\lambda x} \times P(x)}{\int e^{\lambda x} p(x)\,dx}\,dx \qquad \frac{e^{\lambda x} \times P(x)}{\int e^{\lambda x} p(x)\,dx} \longrightarrow Q_\lambda$$

$$\int Q_\lambda = 1$$

$$= \int x^2 Q_\lambda(x)\,dx = \mathbb{E}_{X\sim Q_\lambda}[X^2]$$

$$\psi''(\lambda) = \mathbb{E}_{X\sim Q_\lambda}[X^2] - \mathbb{E}_{X\sim Q_\lambda}[X]^2$$

$$= \mathrm{Var}_{X\sim Q_\lambda}(X) = \mathbb{E}_{X\sim Q_\lambda}\left[\left(X - \mathbb{E}_{X\sim Q_\lambda}[X]\right)^2\right]$$

$$\psi''(\lambda) \leqslant \mathbb{E}_{X\sim Q_\lambda}\left[(X-c)^2\right]$$

$$c \overset{?}{=} \frac{a+b}{2} \leadsto = \mathbb{E}\left[\left(X-\frac{a+b}{2}\right)^2\right]$$

$$\sup_{\lambda} \psi''(\lambda) \leqslant \mathbb{E}\left[\left(b-\frac{a+b}{2}\right)^2\right]$$

$$\left(\frac{b-a}{2}\right)^2$$

$$\psi(\lambda) = \log \mathbb{E}\left[e^{\lambda X}\right] = \mu_X \lambda + \underbrace{\psi''(\eta)}_{} \times \frac{\lambda^2}{2}$$

$$\leqslant \mu\lambda + \frac{(b-a)^2}{4} \times \frac{\lambda^2}{2}$$

$$\mathbb{E}\left[e^{\lambda X}\right] \leqslant e^{\lambda\mu + \frac{\lambda^2 \left(\frac{b-a}{2}\right)^2}{2}} \qquad \left(\frac{b-a}{2}\right)^2 = \sigma^2$$

$$\mathbb{E}[e^{\lambda X}] \le e^{\lambda\mu + \frac{\lambda^2\sigma^2}{2}}$$

(1) $\mathbb{E}[X] = \mu$ (2) $Var(X) \le \sigma^2$

(3) σ^2 tigtest $\longrightarrow$ $Var(X) \overset{?}{=} \sigma^2$

$$\frac{\partial}{\partial\lambda}\mathbb{E}[e^{\lambda X}]\Big|_{\lambda=0} = \mathbb{E}[X]$$

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = f'(x)$$

$$\frac{\partial}{\partial\lambda}\mathbb{E}[e^{\lambda X}] = \lim_{h\to 0}\frac{1}{h}\left(\mathbb{E}[e^{hX}]-1\right)$$

$$\le \lim_{h\to 0}\frac{1}{h}\left(\underbrace{e^{h\mu+\frac{h^2\sigma^2}{2}}}_{g(0+h)} - \underbrace{1}_{g(0)}\right)$$

$$= \frac{\partial}{\partial\lambda} e^{\lambda\mu+\frac{\lambda^2\sigma^2}{2}}\Big|_{\lambda=0} =$$

$$(\mu + \lambda\sigma^2)\, e^{\lambda\mu + \frac{\lambda^2\sigma^2}{2}} \Big|_{\lambda=0} = \mu$$

$$E[X] \leq \mu$$

$$E[X] \geq \mu \longrightarrow = \mu$$

$$\frac{f(0) - f(0-h)}{h}$$

$$\mathbb{E}\left[e^{\lambda(X-\mu)}\right] = 1 + \cancelto{0}{\mathbb{E}[X-\mu]} + \underline{\frac{1}{2}\lambda^2\, \mathbb{E}\left[(X-\mu)^2\right]} + \underline{o(\lambda^2)}$$

$g(\lambda)$

$$\lim_{\lambda \to 0} \frac{g(\lambda)}{\lambda^2} = 0$$

$$\leq e^{\cancel{\lambda\mu} + \frac{\lambda^2\sigma^2}{2}} = 1 + \cancel{\lambda\mu} + \underbrace{\frac{\lambda^2\sigma^2}{2}} + o(\lambda^2)$$

$$e^u = 1 + u + \frac{u^2}{2!} + \cdots$$

$$\cancel{1} + \mathbb{E}\left((X-\mu)^2\right) \times \frac{\lambda^2}{2} + o(\cancel{\lambda^2})$$

$$\leqslant \cancel{1} + \frac{\lambda^2\sigma^2}{2} + o\left(\lambda^2\right)$$

$$E\left((X-\mu)^2\right] \times \frac{\overset{1}{\cancel{\lambda^2}}}{2} \leqslant \frac{\cancel{\lambda^2}\sigma^2}{2} + \overset{o(1)}{\cancel{o(\lambda^2)}}$$

$$\boxed{\lim_{\lambda\to 0} \frac{g(\lambda)}{\lambda} = 0} \qquad \boxed{\mathbb{E}\left[(X-\mu)^2\right] \times \frac{1}{\cancel{2}} \leqslant \frac{\sigma^2}{\cancel{2}}}$$